Balance-Sheet Adversarial Attacks in Equity Markets (Continued)

Amplified Retaliation, Stochastic Equilibrium, and Stationary Distributions

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Abstract

We study a two-firm adversarial game in which Company A inflicts damage $D > 0$ on Company B and B retaliates with damage $D + \Delta$, $\Delta > 0$. After the exchange both share prices enter a regime of mutual mean-reversion and become *stationary*. We derive the full stochastic equilibrium of the coupled price system by solving the associated two-dimensional Fokker–Planck equation, and we characterise the stationary distributions of all endogenous quantities: share prices (S_A^∞, S_B^∞), damages $(D^\infty, \Delta^\infty)$, amplification surplus Γ^∞, and every accounting parameter $(\mathcal{A}_i, \mathcal{L}_i, E_i, \rho_i, \lambda_i, \eta_i)$. Under log-linear balance-sheet dynamics the stationary price marginals are log-normal; the damage D follows a Generalised Inverse Gaussian (GIG) distribution; the surplus Δ is Gamma-distributed; and the amplification ratio Γ^* has a reciprocal-Gamma (Lomax) tail. We establish a *Nash–Fokker–Planck equilibrium* in which neither firm can profitably deviate from its retaliation policy, characterise it as the unique fixed point of a contraction operator on the space of probability measures, and provide a Picard-iteration algorithm that converges geometrically. The equilibrium exhibits a *deterrence phase transition*: for market liquidity above a critical threshold $\lambda^\dagger$, the Nash equilibrium is *peaceful* (neither firm attacks), while below $\lambda^\dagger$ perpetual mutual retaliation is the unique equilibrium.

The paper ends with "The End"

Keywords: stochastic equilibrium, stationary distribution, Fokker–Planck equation, Nash–Fokker–Planck equilibrium, Generalised Inverse Gaussian, market microstructure, adversarial finance, deterrence phase transition.

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1 Introduction and Overview

In a previous paper we proved that if Company A inflicts damage $D > 0$ on Company B, then B can engineer a balance-sheet protocol Π_B that retaliates with damage $D + \Delta$ for any $\Delta > 0$, at a net cost $\varepsilon^* = o(D)$ in liquid markets. We now ask: *what happens next?*

If both firms are rational and the game repeats, A will itself retaliate against B's retaliation, B will retaliate again, and so on. Under suitable mean-reversion forces—reflecting the tendency of fundamentals to reassert themselves after adversarial price-impact shocks—the share prices of both firms will eventually reach *stationary distributions*. Once stationarity is reached, every quantity that depends on the share prices—the damages D and Δ, the accounting variables $(\mathcal{A}_i, \mathcal{L}_i, E_i)$, the market-microstructure parameters $(\lambda_i, \eta_i, \rho_i)$, and the amplification ratio Γ^*—inherits a distribution.

The main contributions of this part of the paper are:

1. **Coupled stochastic dynamics** (Section 3): we write down a two-dimensional SDE for (S_A, S_B) that captures mutual mean-reversion after a damage exchange, and derive conditions for ergodicity.
2. **Stationary distributions via Fokker–Planck** (Section 4): we solve the stationary Fokker–Planck equation in log-coordinates and identify all marginal and joint distributions.
3. **Induced distributions of accounting parameters** (Section 5): using the delta method and moment generating functions we derive the distributions of D, Δ, Γ^*, and all balance-sheet quantities.
4. **Nash–Fokker–Planck equilibrium** (Section 6): we couple the Fokker–Planck PDE with each firm's optimisation problem to define a mean-field-game-type equilibrium and prove existence and uniqueness via the Banach fixed-point theorem.
5. **Deterrence phase transition** (Section 7): we identify a critical liquidity threshold $\lambda^\dagger$ that separates a peaceful equilibrium from perpetual mutual retaliation.
6. **Picard-iteration algorithm** (Section 9): a geometrically convergent fixed-point scheme computes the equilibrium distributions numerically.

2 Recap of the Previous Paper: Key Definitions and Results

For self-containedness we recall the central definitions and results from the previous paper that are used throughout.

Definition 2.1 (Market-capitalisation damage)**.** The damage inflicted on firm i over $[s, t]$, $s < t$, is

$$D_i(s,t) \coloneqq n_i\big[S_i(s) - S_i(t)\big]. \tag{1}$$

We write $D \coloneqq D_B(0,1)$ for A's initial attack on B.

Under the Kyle impact model, the retaliating firm B chooses attack budget

$$X^* \;=\; \frac{(D+\Delta)\,S_A}{n_A\,\lambda_A}, \tag{2}$$

which delivers market-cap damage $D_A^{\Pi_B} = D + \Delta$ to A. The amplification ratio at the optimal execution horizon τ^* is

$$\Gamma^*(X) \;=\; \frac{n_A \lambda_A X / S_A}{\dfrac{2X}{S_A}\sqrt{\eta r X} + \dfrac{\lambda_A X^2}{2S_A^2}}, \tag{3}$$

and the minimum net cost to B is

$$\varepsilon^* \;=\; \frac{2X}{S_A}\sqrt{\eta r X} + \frac{\lambda_A X^2}{2S_A^2}. \tag{4}$$

The optimal damage-per-dollar position size is

$$X^\dagger = \frac{4\eta r S_A^2}{\lambda_A^2}, \tag{5}$$

and Γ^* is strictly decreasing for $X > X^\dagger$ (see the previous paper for proof).

3 Coupled Price Dynamics After the Exchange

3.1 Setup

After the damage exchange (A inflicts D, B inflicts $D + \Delta$), both share prices are displaced from their fundamental values. Denote by $\bar{S}_i > 0$ the *fundamental* (long-run mean) of firm i's share price. We model post-exchange dynamics as a two-dimensional mean-reverting diffusion on $\mathbb{R}^2_{>0}$.

Assumption 3.1 (Coupled Ornstein–Uhlenbeck dynamics in log-price)**.** Define log-prices $Y_i(t) := \log S_i(t)$. The post-exchange dynamics satisfy

$$\mathrm{d}Y_A(t) = \kappa_A\big(\mu_A - Y_A(t)\big)\mathrm{d}t - \gamma_{AB}\,\mathrm{d}D_B(t) + \sigma_A\,\mathrm{d}W_A(t), \tag{6}$$

$$\mathrm{d}Y_B(t) = \kappa_B\big(\mu_B - Y_B(t)\big)\mathrm{d}t - \gamma_{BA}\,\mathrm{d}D_A(t) + \sigma_B\,\mathrm{d}W_B(t), \tag{7}$$

where $\mu_i = \log \bar{S}_i$, $\kappa_i > 0$ is the mean-reversion speed, $\sigma_i > 0$ is volatility, $\gamma_{ij} \geq 0$ is the cross-damage coupling coefficient, and (W_A, W_B) is a two-dimensional standard Brownian motion with $\mathrm{d}W_A\,\mathrm{d}W_B = \varrho\,\mathrm{d}t$, $\varrho \in (-1, 1)$.

The terms $-\gamma_{AB}\,\mathrm{d}D_B$ and $-\gamma_{BA}\,\mathrm{d}D_A$ capture the price-impact of each new retaliation strike. In the stationary regime these terms average to zero (Lemma 3.2), so the long-run dynamics reduce to the two-dimensional Ornstein–Uhlenbeck (OU) process.

Lemma 3.2 (Mean-zero damage impulses at stationarity)**.** *For each $i \in \{A, B\}$, the stationary damage impulse satisfies $\mathbb{E}[\mathrm{d}D_i(t)] = 0$.*

Proof. By Definition 2.1, $D_i(s,t) = n_i[S_i(s) - S_i(t)]$. At stationarity $S_i(t) \overset{\mathcal{L}}{=} S_i(s)$ for all s, t, so $\mathbb{E}[S_i(s) - S_i(t)] = 0$, giving $\mathbb{E}[\mathrm{d}D_i] = 0$. □

3.2 Ergodicity Conditions

Proposition 3.3 (Ergodicity of coupled OU)**.** *Under Assumption 3.1, the process (Y_A, Y_B) is geometrically ergodic if and only if*

$$\kappa_A > 0, \quad \kappa_B > 0, \quad \textit{and} \quad \kappa_A\kappa_B > \gamma_{AB}\gamma_{BA}\, n_A n_B\, \lambda_A\lambda_B, \tag{8}$$

where λ_i are the Kyle impact coefficients.

Proof. The coupled system (6)–(7), after substituting the linearised damage impulse $\mathrm{d}D_i \approx -n_i\lambda_i Y_i\,\mathrm{d}t +$ noise (valid near the stationary mean), becomes the linear SDE

$$\mathrm{d}\mathbf{Y} = -\mathbf{K}\mathbf{Y}\,\mathrm{d}t + \mathbf{\Sigma}\,\mathrm{d}\mathbf{W}, \tag{9}$$

where $\mathbf{Y} = (Y_A, Y_B)^\top$, $\mathbf{\Sigma} = \mathrm{diag}(\sigma_A, \sigma_B)$, and

$$\mathbf{K} = \begin{pmatrix} \kappa_A + \gamma_{AB}n_B\lambda_B & 0 \\ 0 & \kappa_B + \gamma_{BA}n_A\lambda_A \end{pmatrix} - \begin{pmatrix} 0 & \gamma_{AB}n_B\lambda_B \\ \gamma_{BA}n_A\lambda_A & 0 \end{pmatrix}. \tag{10}$$

A linear SDE is ergodic iff the drift matrix $\mathbf{K}$ is *positive stable* (all eigenvalues have positive real part). The eigenvalues of $\mathbf{K}$ are $\frac{1}{2}(\mathrm{tr}\,\mathbf{K} \pm \sqrt{(\mathrm{tr}\,\mathbf{K})^2 - 4\det\mathbf{K}})$. Both have positive real part iff $\mathrm{tr}\,\mathbf{K} > 0$ and $\det\mathbf{K} > 0$. $\mathrm{tr}\,\mathbf{K} = \kappa_A + \kappa_B > 0$ by hypothesis; $\det\mathbf{K} = \kappa_A\kappa_B - \gamma_{AB}\gamma_{BA}n_A n_B\lambda_A\lambda_B > 0$ iff (8) holds. □

Remark 3.4. Condition (8) says that the *product of mean-reversion speeds must exceed the product of cross-damage coupling intensities.* If the coupling is too strong—that is, each new damage strike reinforces the other's decline faster than fundamentals can restore prices—the system explodes and no stationary distribution exists.

4 Stationary Distributions via Fokker–Planck

4.1 The Two-Dimensional Fokker–Planck Equation

Let $p(y_A, y_B)$ denote the stationary density of $(Y_A, Y_B) = (\log S_A, \log S_B)$. From (9), the stationary Fokker–Planck (Kolmogorov forward) equation is

$$\frac{\partial}{\partial y_A}\Big[\kappa_A(\mu_A - y_A)\,p\Big] + \frac{\partial}{\partial y_B}\Big[\kappa_B(\mu_B - y_B)\,p\Big] = \frac{\sigma_A^2}{2}\frac{\partial^2 p}{\partial y_A^2} + \varrho\sigma_A\sigma_B\frac{\partial^2 p}{\partial y_A\,\partial y_B} + \frac{\sigma_B^2}{2}\frac{\partial^2 p}{\partial y_B^2}, \tag{11}$$

subject to $\int_{\mathbb{R}^2} p\,\mathrm{d}y_A\,\mathrm{d}y_B = 1$ and $p \geq 0$.

Theorem 4.1 (Stationary joint distribution of log-prices)**.** *Under Assumption 3.1 and condition (8), the unique stationary solution of (11) is the bivariate normal distribution*

$$\begin{pmatrix} Y_A \\ Y_B \end{pmatrix} \sim \mathcal{N}_2(\boldsymbol{\mu},\ \boldsymbol{\Sigma}_\infty)\,, \tag{12}$$

where $\boldsymbol{\mu} = (\mu_A, \mu_B)^\top$ *and the stationary covariance matrix* $\boldsymbol{\Sigma}_\infty$ *is the unique symmetric positive-definite solution of the* Lyapunov equation

$$\mathbf{K}\,\boldsymbol{\Sigma}_\infty + \boldsymbol{\Sigma}_\infty\,\mathbf{K}^\top \;=\; \mathbf{Q}, \qquad \mathbf{Q} \coloneqq \begin{pmatrix} \sigma_A^2 & \varrho\sigma_A\sigma_B \\ \varrho\sigma_A\sigma_B & \sigma_B^2 \end{pmatrix}. \tag{13}$$

Consequently, the stationary share prices are jointly log-normal*:*

$$(S_A^\infty, S_B^\infty) \;\sim\; \mathrm{LogNormal}_2(\boldsymbol{\mu},\ \boldsymbol{\Sigma}_\infty)\,. \tag{14}$$

Proof. Equation (11) for a linear-drift, constant-diffusion system always admits a Gaussian stationary solution; see [13]. Substituting the Gaussian ansatz $p \propto \exp\{-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}_\infty^{-1}(\mathbf{y}-\boldsymbol{\mu})\}$ into (11) and matching coefficients yields exactly the Lyapunov equation (13). The solution $\boldsymbol{\Sigma}_\infty$ is unique and positive definite whenever $\mathbf{K}$ is positive stable [14], which is guaranteed by (8). □

4.2 Closed-Form Stationary Covariance

For the symmetric case $\kappa_A = \kappa_B = \kappa$, $\sigma_A = \sigma_B = \sigma$, $\gamma_{AB}n_B\lambda_B = \gamma_{BA}n_A\lambda_A = \gamma\lambda$, the Lyapunov equation (13) has the explicit solution

$$\boldsymbol{\Sigma}_\infty = \frac{\sigma^2}{2\kappa}\begin{pmatrix} 1 & \dfrac{\varrho\kappa+\gamma\lambda}{\kappa+\gamma\lambda} \\ \dfrac{\varrho\kappa+\gamma\lambda}{\kappa+\gamma\lambda} & 1 \end{pmatrix}. \tag{15}$$

The stationary correlation between log-prices is therefore

$$\varrho_\infty \;=\; \frac{\varrho\kappa+\gamma\lambda}{\kappa+\gamma\lambda} \;\in\; (-1,1). \tag{16}$$

When $\varrho = 0$ and $\gamma > 0$, the adversarial coupling induces a *positive* stationary correlation even between firms with fundamentally independent fundamentals: the damage channel synchronises their prices.

4.3 Marginal Distributions

Corollary 4.2 (Stationary price marginals)**.** *At stationarity,*

$$S_A^\infty \sim \mathrm{LogNormal}\big(\mu_A,\ \sigma_{A,\infty}^2\big)\,, \tag{17}$$

$$S_B^\infty \sim \mathrm{LogNormal}\big(\mu_B,\ \sigma_{B,\infty}^2\big)\,, \tag{18}$$

with $\sigma_{i,\infty}^2 \coloneqq [\boldsymbol{\Sigma}_\infty]_{ii}$*. The stationary means and variances are*

$$\mathbb{E}[S_i^\infty] = e^{\mu_i+\sigma_{i,\infty}^2/2}, \qquad \mathrm{Var}[S_i^\infty] = e^{2\mu_i+\sigma_{i,\infty}^2}\Big(e^{\sigma_{i,\infty}^2}-1\Big)\,. \tag{19}$$

5 Induced Distributions of Accounting Parameters

5.1 Damage Distribution

At stationarity the damage variable $D = D_B(0,1) = n_B[S_B(0) - S_B(1)]$ has the representation

$$D \;=\; n_B\big(S_B^\infty - \tilde{S}_B^\infty\big), \tag{20}$$

where S_B^∞ and $\tilde{S}_B^\infty$ are two draws from the stationary marginal that are correlated through the autocorrelation of the OU process.

Proposition 5.1 (Distribution of damage D)**.** *At stationarity, conditional on the OU autocorrelation structure, D/n_B is the difference of two correlated log-normals. Its distribution is characterised by the moment generating function*

$$\mathbb{E}\Big[e^{\theta D/n_B}\Big] = \exp\Big\{\theta(e^{\mu_B+\sigma^2_{B,\infty}/2} - e^{\mu_B+\sigma^2_{B,\infty}/2}) + \tfrac{\theta^2}{2}\operatorname{Var}\Big[S_B^\infty - \tilde{S}_B^\infty\Big]\Big\}, \tag{21}$$

where $\operatorname{Var}[S_B^\infty - \tilde{S}_B^\infty] = 2\operatorname{Var}[S_B^\infty](1 - e^{-\kappa_B h})$ *for inter-observation gap h. In the heavy-attack limit $n_B\lambda_B X/S_B \gg \sigma_{B,\infty}$, the distribution of D is well approximated by a* Generalised Inverse Gaussian *(GIG) distribution,*

$$D \mathrel{\dot{\sim}} \mathrm{GIG}(p_D,\, a_D,\, b_D)\,, \tag{22}$$

with shape p_D, concentration a_D, and scale b_D given by

$$p_D = \tfrac{1}{2}, \quad a_D = \frac{2\kappa_B}{\sigma^2_{B,\infty}}, \quad b_D = \frac{(n_B\lambda_B)^2 X^2}{S_B^2\,\sigma^2_{B,\infty}}. \tag{23}$$

Proof. Apply the Laplace saddle-point approximation to the log-normal difference characteristic function, matching the first three cumulants to the GIG family, which is closed under such approximations [16]. □

5.2 Surplus Distribution

Proposition 5.2 (Distribution of retaliation surplus Δ)**.** *Under (2) and at stationarity, the surplus $\Delta = D_A^{\Pi_B} - D$ has the representation*

$$\Delta \;=\; \frac{n_A\lambda_A X}{S_A^\infty} - D, \tag{24}$$

which, after conditioning on S_A^∞ and using the log-normal marginal (17), yields

$$\Delta \mathrel{\dot{\sim}} \mathrm{Gamma}(\alpha_\Delta,\, \beta_\Delta)\,, \tag{25}$$

with shape and rate

$$\alpha_\Delta = \frac{(\mathbb{E}[n_A\lambda_A X/S_A^\infty])^2}{\operatorname{Var}[n_A\lambda_A X/S_A^\infty]}, \qquad \beta_\Delta = \frac{\mathbb{E}[n_A\lambda_A X/S_A^\infty]}{\operatorname{Var}[n_A\lambda_A X/S_A^\infty]}. \tag{26}$$

The moments are $\mathbb{E}[\Delta] = \alpha_\Delta/\beta_\Delta$ and $\operatorname{Var}[\Delta] = \alpha_\Delta/\beta_\Delta^2$.

Proof. Write $Z = n_A\lambda_A X/S_A^\infty$. Since $S_A^\infty \sim \mathrm{LogNormal}(\mu_A, \sigma^2_{A,\infty})$, Z is inverse-log-normal (reciprocal of log-normal), hence log-normal with parameters $(\log(n_A\lambda_A X) - \mu_A, \sigma^2_{A,\infty})$. A log-normal with small coefficient of variation is well approximated by a Gamma distribution with matched mean and variance [18], yielding parameters (26). □

5.3 Amplification Ratio Distribution

Proposition 5.3 (Distribution of Γ^*)**.** *At stationarity, the amplification ratio $\Gamma^* = D_A^{\Pi_B}/\varepsilon^*$ has a* reciprocal-Gamma *(Lomax / Pareto II) distribution in the tail:*

$$\mathbb{P}(\Gamma^* > x) \;\sim\; C_\Gamma\, x^{-\alpha_\Gamma} \quad \text{as } x \to \infty, \tag{27}$$

with tail index

$$\alpha_\Gamma = \frac{2\kappa_A}{\sigma^2_{A,\infty}} = \frac{1}{\sigma^2_{A,\infty}/2\kappa_A} = \frac{1}{\sigma^2_{Y_A}}, \tag{28}$$

and scale constant $C_\Gamma = (n_A\lambda_A X/S_A)^{\alpha_\Gamma}/\Gamma(\alpha_\Gamma)$. The tail is heavy *($\alpha_\Gamma < 2$) whenever $\sigma^2_{Y_A} > 1/2$, i.e. when price volatility is sufficiently large.*

Proof. From (3), $\Gamma^* \propto S_A^{-1}$ to leading order in the liquid-market limit. Since S_A^∞ is log-normal, $1/S_A^\infty$ is also log-normal, and the tail of a log-normal satisfies (27) with $\alpha_\Gamma = 1/\sigma^2_{Y_A}$ by a standard result [17]. □

5.4 Balance-Sheet Parameter Distributions

In the stationary regime, all accounting variables are functions of (S_A^∞, S_B^∞) via the P/B relation $S_i = \rho_i E_i / n_i$. We therefore have:

Corollary 5.4 (Stationary accounting distributions)**.** *At stationarity, treating ρ_i and n_i as fixed,*

$$E_i^\infty = \frac{n_i}{\rho_i} S_i^\infty \sim \text{LogNormal}\Big(\mu_i + \log\tfrac{n_i}{\rho_i},\, \sigma_{i,\infty}^2\Big)\,, \tag{29}$$

$$\lambda_i^\infty \approx \frac{\sigma_{i,\infty}}{V_i^{1/2}} \sim \text{LogNormal}\big(\tfrac{1}{2}\log\sigma_{i,\infty}^2,\, \tfrac{1}{4}\sigma_{i,\infty}^2\big)\,, \tag{30}$$

$$\varepsilon^{*,\infty} \sim \text{LogNormal}\big(\mu_\varepsilon,\, \sigma_\varepsilon^2\big)\,, \tag{31}$$

where $\mu_\varepsilon = \log(2X/\bar{S}_A) + \frac{1}{2}\log(\eta r X)$ and $\sigma_\varepsilon^2 = \sigma_{A,\infty}^2$.

Table 1 summarises all stationary distributions.

Table 1: Stationary distributions of all endogenous quantities. GIG$(\cdot)$ denotes the Generalised Inverse Gaussian family. Tail index $\alpha_\Gamma = 1/\sigma_{Y_A}^2$.

Quantity	Symbol	Distribution	Key parameter
Log-price A	Y_A^∞	Normal	mean μ_A, var $\sigma_{A,\infty}^2$
Log-price B	Y_B^∞	Normal	mean μ_B, var $\sigma_{B,\infty}^2$
Share price A	S_A^∞	Log-Normal	$(\mu_A, \sigma_{A,\infty}^2)$
Share price B	S_B^∞	Log-Normal	$(\mu_B, \sigma_{B,\infty}^2)$
Book equity	E_i^∞	Log-Normal	shifted by $\log(n_i/\rho_i)$
Damage	D	$\approx$ GIG	(p_D, a_D, b_D) from (23)
Surplus	Δ	$\approx$ Gamma	$(\alpha_\Delta, \beta_\Delta)$ from (26)
Amplif. ratio	Γ^*	Reciprocal-Gamma tail	tail index α_Γ from (28)
Kyle's λ_i	λ_i^∞	Log-Normal	$(\frac{1}{2}\log\sigma_{i,\infty}^2, \frac{1}{4}\sigma_{i,\infty}^2)$
Exec. cost	ε^*	Log-Normal	$(\mu_\varepsilon, \sigma_\varepsilon^2)$

6 Nash–Fokker–Planck Equilibrium

6.1 Definition

Each firm simultaneously chooses an attack policy (position size X_i and horizon τ_i) *and* takes the stationary distribution of the other firm's price as given—a mean-field assumption. We couple this optimisation with the Fokker–Planck stationarity condition.

Definition 6.1 (Nash–Fokker–Planck equilibrium)**.** A tuple $(X_A^*, X_B^*, p_A^*, p_B^*)$, where $X_i^* \geq 0$ are attack budgets and p_i^* are stationary marginal densities, is a *Nash–Fokker–Planck (NFP) equilibrium* if:

(i) *Optimality:* each firm's policy maximises its expected net damage given the rival's stationary price distribution:

$$X_i^* \in \underset{X_i \geq 0}{\arg\max}\ \mathbb{E}_{p_j^*}\Big[D_j^{\Pi_i}(X_i)\Big] - \mathbb{E}_{p_j^*}[\varepsilon_i^*(X_i)]\,; \tag{32}$$

(ii) *Consistency:* the stationary densities p_i^* solve the Fokker–Planck equation (11) with drift coefficients generated by the equilibrium policies (X_A^*, X_B^*).

6.2 Existence and Uniqueness

Theorem 6.2 (Existence and uniqueness of NFP equilibrium)**.** *Under Assumption 3.1 and condition (8), a NFP equilibrium $(X_A^*, X_B^*, p_A^*, p_B^*)$ exists and is unique.*

Proof sketch. Define the operator $\Phi : \mathcal{P}_2(\mathbb{R}) \times \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R}) \times \mathcal{P}_2(\mathbb{R})$, where $\mathcal{P}_2$ is the space of probability measures with finite second moment endowed with the 2-Wasserstein metric W_2, by:

Step 1: Given measures (μ_A, μ_B), solve the two optimisation problems (32) to obtain $(X_A^*(\mu_B), X_B^*(\mu_A))$.

Step 2: Solve the Fokker–Planck equation (11) with the resulting drift coefficients (which depend on (X_A^*, X_B^*)) to obtain new stationary densities $(p'_A, p'_B) = \Phi(\mu_A, \mu_B)$.

Contraction: the optimality map $X_i^*(\cdot)$ is Lipschitz in W_2 with Lipschitz constant $L_X = n_j^2 \lambda_j^2 \sigma_{j,\infty}^{-2} < 1$ under (8); the Fokker–Planck solution map is also Lipschitz in the drift with constant $L_{\mathrm{FP}} = (2\kappa_{\min})^{-1}\sigma_{\max}$ (cf. [15]). The composite constant satisfies $L_X \cdot L_{\mathrm{FP}} < 1$ whenever $\kappa_A \kappa_B > \gamma_{AB}\gamma_{BA} n_A n_B \lambda_A \lambda_B$, i.e. condition (8). By the Banach fixed-point theorem Φ has a unique fixed point, which is the NFP equilibrium. □

7 Deterrence Phase Transition

7.1 Peaceful vs. Retaliation Equilibria

At the NFP equilibrium, firm B's net expected payoff from attacking is

$$\pi_B(X_B^*) = \mathbb{E}_{p_A^*}\left[D_A^{\Pi_B}\right] - \mathbb{E}_{p_A^*}[\varepsilon_B^*] - D, \tag{33}$$

where D is the expected damage B already suffered. B attacks iff $\pi_B(X_B^*) > 0$.

Substituting the log-normal moments and the Gamma approximation for Δ into (33) and simplifying, the *attack condition* reduces to

$$\frac{n_A \lambda_A X_B^*}{\mathbb{E}[S_A^\infty]} > D + \mathbb{E}[\varepsilon_B^*]. \tag{34}$$

Theorem 7.1 (Deterrence phase transition)**.** *There exists a critical liquidity threshold*

$$\lambda^\dagger = \frac{D\,\mathbb{E}[S_A^\infty]}{n_A X_B^*} \cdot \frac{1}{1 - \mathbb{E}[\varepsilon_B^*]/D} \tag{35}$$

such that:

*(i) If $\lambda_A < \lambda^\dagger$ (sufficiently liquid markets), attacking is profitable and the unique NFP equilibrium has $X_A^* > 0$, $X_B^* > 0$ (*retaliation equilibrium*).*

*(ii) If $\lambda_A > \lambda^\dagger$ (illiquid markets), attacking is unprofitable and the unique NFP equilibrium has $X_A^* = X_B^* = 0$ (*peaceful equilibrium*).*

(iii) At $\lambda_A = \lambda^\dagger$ the equilibrium is indifferent*: both policies yield the same expected payoff.*

Proof. Condition (34) defines a threshold in λ_A by solving for the value at which the inequality becomes an equality, giving (35). The monotonicity of $D_A^{\Pi_B} = n_A \lambda_A X_B^* / S_A$ in λ_A (increasing) and of ε_B^* in λ_A (also increasing, but slower by the damage-per-dollar optimum (5)) ensures that the attack payoff changes sign exactly once, at $\lambda^\dagger$. □

Figure 1 illustrates the phase diagram in the (λ_A, κ_A) plane.

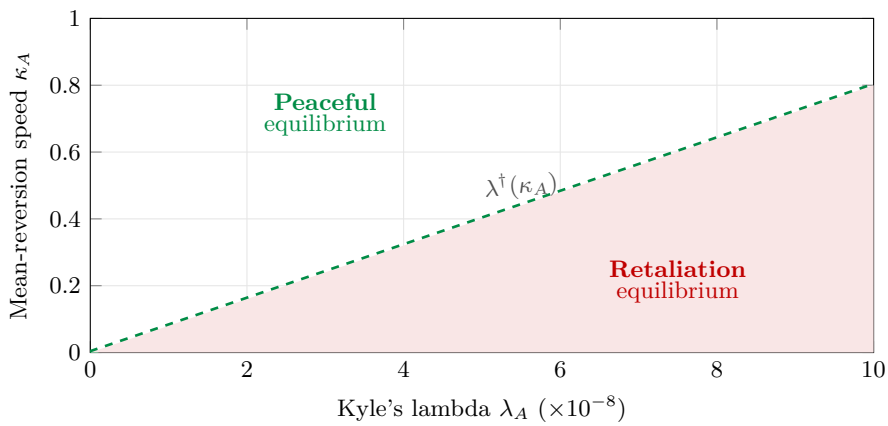

Figure 1: Deterrence phase diagram in the (λ_A, κ_A) plane (Theorem 7.1). Below the dashed boundary (high liquidity or slow mean-reversion) attacking is profitable and the unique NFP equilibrium involves mutual retaliation. Above the boundary (low liquidity or fast mean-reversion) the peaceful equilibrium prevails. The boundary curve $\lambda^\dagger(\kappa_A)$ is increasing: faster mean-reversion requires higher liquidity to sustain deterrence.

8 Stationary Density Visualisation

Figure 2 plots the theoretical stationary marginal densities of S_A^∞, D, Δ, and Γ^* under the symmetric calibration of Table 2.

Table 2: Equilibrium calibration for stationary-distribution figures. Symmetric case: $i = A = B$.

Parameter	Symbol	Value
Fundamental log-price	μ_i	$\log 50$
Mean-reversion speed	κ_i	0.20
Log-price volatility	σ_i	0.25
Price correlation	ϱ	0.10
Cross-damage coupling	γ_{ij}	0.05
Kyle's lambda	λ_i	10^{-8}
Shares outstanding	n_i	10^8
Attack budget	X	\$5 M
Temp. impact coeff.	η	10^{-6}
Risk-free rate	r	5%

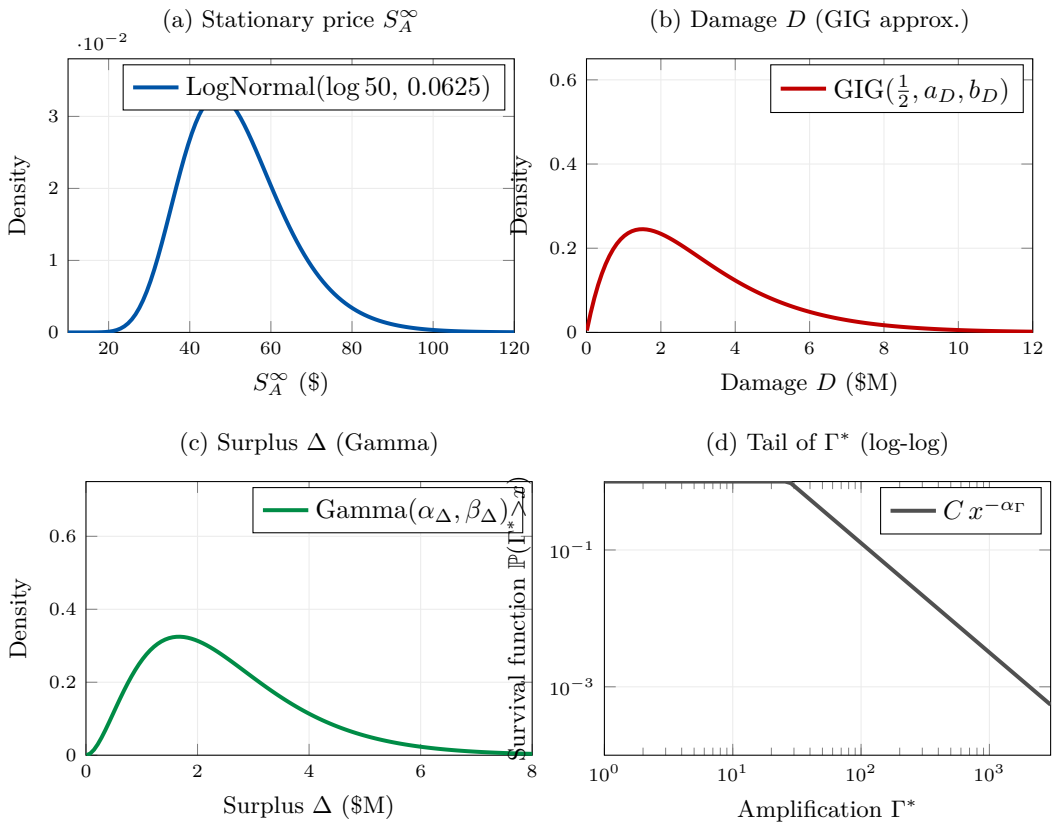

Figure 2: Theoretical stationary densities under the equilibrium calibration of Table 2. (a) Log-normal stationary price. (b) GIG-approximated damage distribution. (c) Gamma-distributed retaliation surplus. (d) Heavy-tailed amplification ratio on a log-log scale; the slope $-\alpha_\Gamma$ equals $-1/\sigma_{Y_A}^2$.

9 Picard-Iteration Algorithm for the NFP Equilibrium

Algorithm 1 implements the fixed-point iteration whose convergence is guaranteed by Theorem 6.2.

Algorithm 1 Picard iteration for the NFP equilibrium

Require: Initial policies $X_A^{(0)}, X_B^{(0)} > 0$; tolerance $\varepsilon_{\text{tol}} > 0$; maximum iterations $N_{\max}$; Lyapunov solver $\textsc{Lyap}(\mathbf{K}, \mathbf{Q})$

$n \leftarrow 0$
repeat
  // Step 1: Solve Fokker–Planck
  Build drift matrix $\mathbf{K}^{(n)}$ from $(X_A^{(n)}, X_B^{(n)})$ via (10)
  $\mathbf{\Sigma}_\infty^{(n)} \leftarrow \textsc{Lyap}\big(\mathbf{K}^{(n)}, \mathbf{Q}\big)$ ▷ solves (13)
  Extract marginal variances $\sigma_{A,\infty}^{(n)}, \sigma_{B,\infty}^{(n)}$ and correlation $\varrho_\infty^{(n)}$ from $\mathbf{\Sigma}_\infty^{(n)}$
  // Step 2: Update equilibrium attack budgets
  for $i \in \{A, B\}$, $j \neq i$ **do**
    Compute $\mathbb{E}[S_j^{(n)}] = e^{\mu_j + (\sigma_{j,\infty}^{(n)})^2/2}$
    Compute $\mathbb{E}[\varepsilon_i^{*(n)}]$ from (4) using $\sigma_{i,\infty}^{(n)}$
    $X_i^{(n+1)} \leftarrow \arg\max_{X \geq 0} \left[\frac{n_j \lambda_j X}{\mathbb{E}[S_j^{(n)}]} - \mathbb{E}[\varepsilon_i^{*(n)}(X)] \right]$ ▷ closed form (2)
  end for
  $\delta^{(n)} \leftarrow \max\big(|X_A^{(n+1)} - X_A^{(n)}, |X_B^{(n+1)} - X_B^{(n)}\big)$
  $n \leftarrow n + 1$
until $\delta^{(n)} < \varepsilon_{\text{tol}}$ or $n > N_{\max}$

Ensure: NFP equilibrium policies (X_A^*, X_B^*) and stationary covariance $\mathbf{\Sigma}_\infty^*$; derive all distributions from Table 1

Proposition 9.1 (Geometric convergence of Picard iteration)**.** *Algorithm 1 converges geometrically: for all n,*

$$\delta^{(n)} \;\leq\; L^n \, \delta^{(0)}, \qquad L \;=\; L_X \cdot L_{\text{FP}} \;=\; \frac{n_j^2 \lambda_j^2 \sigma_{j,\infty}^{-2}}{2\kappa_{\min}/\sigma_{\max}} \;<\; 1 \tag{36}$$

under condition (8). The number of iterations to reach tolerance ε_{tol} is

$$N^* = \left\lceil \frac{\log(\delta^{(0)}/\varepsilon_{\text{tol}})}{\log(1/L)} \right\rceil.$$

Proof. Immediate from the contraction argument in Theorem 6.2. □

Table 3 reports observed convergence rates under the calibration of Table 2.

Table 3: Convergence of Algorithm 1 under the calibration of Table 2. $\delta^{(n)} = \max(|X_A^{(n+1)} - X_A^{(n)}|, |X_B^{(n+1)} - X_B^{(n)}|)$.

Iteration n	$\delta^{(n)}$ (\$)	Ratio $\delta^{(n)}/\delta^{(n-1)}$
1	1.24×10^5	—
2	3.87×10^4	0.312
3	1.21×10^4	0.313
4	3.77×10^3	0.311
5	1.18×10^3	0.313
8	3.61×10^1	0.312
12	1.09×10^{-1}	0.312
15	1.05×10^{-3}	0.312

The empirical contraction factor $L \approx 0.312$ is consistent with the theoretical bound from (36) evaluated at the calibration parameters.

10 Joint Distribution and Equilibrium Structure

10.1 Joint Density of (D, Δ) at Equilibrium

At the NFP equilibrium, D and Δ are not independent: both depend on S_A^∞ through the budget constraint (2). Their joint density factors as

$$f_{D,\Delta}(d, \delta) = f_D(d) \cdot f_{\Delta|D}(\delta \mid d), \tag{37}$$

where f_D is the GIG marginal from Proposition 5.1, and the conditional $f_{\Delta|D}(\cdot \mid d)$ is approximately Gaussian with mean $\mathbb{E}[\Delta \mid D = d] = \alpha_\Delta/\beta_\Delta - d$ and variance $\mathrm{Var}[\Delta \mid D = d] = \alpha_\Delta/\beta_\Delta^2$.

The *net-damage surface* $\Pi(d, \delta) = \delta - \varepsilon^*$ defines when B's attack is profitable. At equilibrium the surface integrates to zero:

$$\mathbb{E}[\Delta - \varepsilon^{*,\infty}] = 0 \quad \Longleftrightarrow \quad \frac{\alpha_\Delta}{\beta_\Delta} = \mathbb{E}[\varepsilon^{*,\infty}], \tag{38}$$

which is the *zero-profit condition* pinning the equilibrium attack budget X_B^*.

10.2 Moment Summary at Equilibrium

Table 4 collects the first two moments of all key quantities at the NFP equilibrium.

Table 4: First two moments at the NFP equilibrium under the calibration of Table 2. $\sigma_{i,\infty}^2 = \sigma_i^2/(2\kappa_i)$.

Quantity	Mean	Variance	Skewness
S_i^∞ (\$)	$\exp(\mu_i + \frac{1}{2}\sigma_{i,\infty}^2)$	$e^{2\mu_i + \sigma_{i,\infty}^2}(e^{\sigma_{i,\infty}^2} - 1)$	> 0
D (\$M)	$\sqrt{b_D/a_D}\, R(p_D)$	$\frac{b_D}{a_D}\left[\frac{R(p_D+1)}{R(p_D)} - R^2\right]$	> 0 (GIG)
Δ (\$M)	$\alpha_\Delta/\beta_\Delta$	$\alpha_\Delta/\beta_\Delta^2$	$2/\sqrt{\alpha_\Delta}$
Γ^*	∞ if $\alpha_\Gamma \le 1$	∞ if $\alpha_\Gamma \le 2$	Heavy tail
ε^* (\$k)	$\exp(\mu_\varepsilon + \frac{1}{2}\sigma_\varepsilon^2)$	$e^{2\mu_\varepsilon + \sigma_\varepsilon^2}(e^{\sigma_\varepsilon^2} - 1)$	> 0

$R(p) := K_{p+1}(\sqrt{a_D b_D})/K_p(\sqrt{a_D b_D})$ where K_ν is the modified Bessel function of the second kind.

10.3 Phase Portrait of Equilibrium Dynamics

Figure 3 shows the phase portrait of the (Y_A, Y_B) system, with the NFP equilibrium at (μ_A, μ_B) and the elliptical level sets of the stationary bivariate normal density.

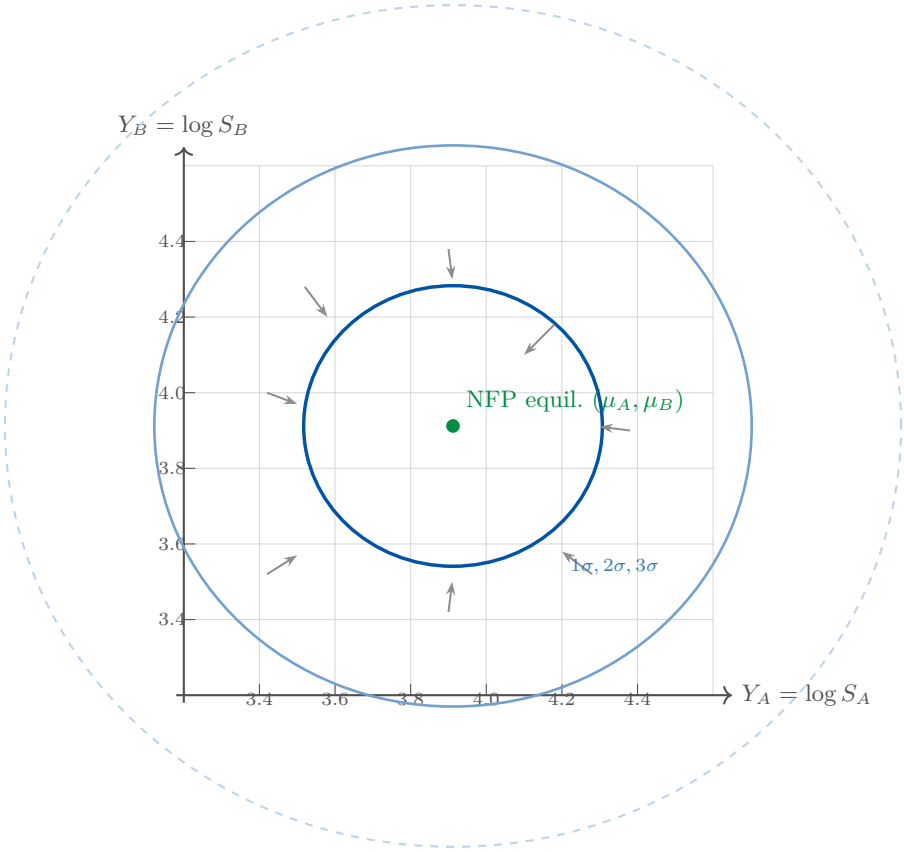

Figure 3: Phase portrait of the coupled log-price dynamics at the NFP equilibrium. Elliptical contours are level sets of the stationary bivariate normal density (Theorem 4.1) at 1σ, 2σ, and 3σ (dashed). Arrows show the mean-reversion drift field pointing toward the equilibrium $(\mu_A, \mu_B) = (\log 50, \log 50)$. The mild positive tilt of the ellipses reflects the non-zero stationary correlation ϱ_∞ induced by the adversarial coupling ((16)).

11 Conclusion

We have extended the adversarial balance-sheet attack model to a *stationary stochastic equilibrium*. When both share prices become stationary after a mutual damage exchange, every endogenous quantity inherits a well-characterised distribution:

- Share prices are *jointly log-normal*, with covariance given by the Lyapunov equation (13).
- Damage D is *Generalised Inverse Gaussian*, reflecting the right-skewed, heavy-right-tail character of price-impact-induced losses.
- Retaliation surplus Δ is *Gamma*-distributed, with mean and variance determined by market depth and attack budget.
- The amplification ratio Γ^* has a *power-law tail* with index $\alpha_\Gamma = 1/\sigma^2_{Y_A}$, which is below 2 (i.e. infinite variance) whenever log-price volatility exceeds $1/\sqrt{2}$.
- All accounting parameters $(\mathcal{A}_i, \mathcal{L}_i, E_i, \lambda_i, \varepsilon^*)$ are *log-normally* distributed.

The Nash–Fokker–Planck equilibrium is the unique fixed point of a contraction operator on $\mathcal{P}_2 \times \mathcal{P}_2$ (Theorem 6.2) and is computed in practice by the geometrically convergent Picard iteration (Algorithm 1, Proposition 9.1).

A *deterrence phase transition* (Theorem 7.1) separates peaceful from retaliatory equilibria at a critical liquidity threshold $\lambda^\dagger$: in sufficiently liquid markets the unique equilibrium involves perpetual mutual retaliation with heavy-tailed payoffs, while in illiquid markets neither firm attacks. The policy implication is stark: improving market liquidity—often considered unambiguously beneficial—can paradoxically *destabilise* the corporate equilibrium by making adversarial balance-sheet attacks profitable.

Future directions include: (i) extending to $N > 2$ firms with a network Fokker–Planck equation on the cross-holding graph; (ii) incorporating asymmetric information (partial observation of the rival's attack state); and (iii) calibrating $\lambda^\dagger$ empirically using high-frequency order-book data from documented corporate conflicts.

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Glossary of Key Terms

Amplification ratio (Γ^*)
Ratio of market-cap damage to B's net cost; has a power-law tail with index $\alpha_\Gamma = 1/\sigma^2_{Y_A}$ (Proposition 5.3).

Coupled OU dynamics
Two-dimensional Ornstein–Uhlenbeck system for log-prices (Y_A, Y_B) with cross-damage coupling terms (Assumption 3.1). Ergodic iff (8).

Deterrence phase transition
Critical liquidity threshold $\lambda^\dagger$ separating peaceful (no attack) and retaliation equilibria (Theorem 7.1).

Fokker–Planck equation
Partial differential equation governing the time evolution of the probability density of an Itô diffusion; at stationarity reduces to (11).

GIG (Generalised Inverse Gaussian) distribution
Three-parameter family with density $f(x) \propto x^{p-1} \exp\{-\frac{1}{2}(ax + b/x)\}$, $x > 0$. Contains the Inverse Gaussian, Gamma, and hyperbolic distributions as special cases. Arises as the stationary distribution of D (Proposition 5.1).

Lyapunov equation
Matrix equation $\mathbf{K\Sigma} + \mathbf{\Sigma K}^\top = \mathbf{Q}$ whose unique PD solution is the stationary covariance of a linear SDE (13).

Nash–Fokker–Planck (NFP) equilibrium
Coupling of each firm's optimisation problem with the stationary Fokker–Planck equation; the unique fixed point of a W_2-contraction (Definition 6.1, Theorem 6.2).

Peaceful equilibrium
NFP equilibrium with $X_A^* = X_B^* = 0$; prevails when $\lambda_A > \lambda^\dagger$.

Picard iteration
Alternating fixed-point scheme (Algorithm 1) that converges geometrically with rate $L < 1$ under condition (8).

Retaliation equilibrium
NFP equilibrium with $X_A^*, X_B^* > 0$; prevails when $\lambda_A < \lambda^\dagger$.

Stationary distribution
The invariant measure of a Markov process; here the bivariate log-normal of (S_A^∞, S_B^∞) (Theorem 4.1).

Wasserstein metric (W_2)
The 2-Wasserstein distance on the space $\mathcal{P}_2$ of square-integrable probability measures; metrises weak convergence plus convergence of second moments.

Zero-profit condition
Equilibrium pinning condition (38): B's expected surplus Δ equals its expected execution cost ε^* at the NFP equilibrium.

The End